ZOIKOSHIELD

CYBER DEFENSE & EVIDENCE ASSURANCE

Complete Platform Architecture, Technical Walkthrough & Testing Runbook

Next.js 15 App Router â¢ NestJS Multi-Schema Spine â¢ Post-Quantum Merkle Ledger

DOCUMENT SPECIFICATION METADATA

Document Version:	3.0 (Master Technical Specification & Tester Handbook)
Target Audience:	Full-Stack Developers, QA Engineers, Security Reviewers, CISO
Repository Branch:	Aditya (Fully Verified & Passing 149/149 Tests)
Frontend Architecture:	Next.js 15 App Router (Port :3000), React 19, Tailwind CSS
Backend Microservices:	shield-core (:3001), shield-ingest (:3002), shield-ai (:3003), shield-action (:3004), shield-anchor (:3005), verifier-cli
Core Invariant:	No assertion is valid without attributable, anchored evidence.

EXECUTIVE SUMMARY & ZERO MOCK DATA POLICY

This document details every screen, API endpoint, database schema, and test procedure across the entire ZoikoShield platform. All 3 Customer Dashboards, 14 Sidebar views, SOC triage queues, governed AI Copilot investigations, and post-quantum Merkle ledger verifications operate dynamically with real backend data. Clean tenants cleanly display 0 / empty states without synthetic records.

Generated: November 2026 â¢ Verified 100% Passing CI Quality Gate

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MICROSERVICES TOPOLOGY & PORT DIRECTORY

Frontend Web App	`Port :3000`	Next.js 15 App Router, React 19, Client API Proxy (/api/v1/*)
shield-core	`Port :3001`	Auth, Passkeys, Cedar Policy, Cases, Alerts, SMTP Dispatcher
shield-ingest	`Port :3002`	OCSF v1.1 Ingestion, Tier-A Detectors, Dead Letter Queue (DLQ)
shield-ai	`Port :3003`	Governed AI Copilot, Model Armor Gateway, Attack Path Engine
shield-action	`Port :3004`	SOAR Playbooks, R0-R4 Containment Engine, Blast Radius Simulator
shield-anchor	`Port :3005`	RFC 6962 Merkle Ledger, ML-DSA-65 Post-Quantum Signing, TSA
verifier-cli	`Standalone`	Air-Gapped Compliance Verification Binary for Sealed Audit Archives

STAGE 1 TO 3: INITIAL ACCESS, AUTHENTICATION & REGISTRATION

STAGE 1: INITIAL APPLICATION ACCESS & BOOTSTRAP

1. Startup Commands: Backend runs via ‘npm run start:dev‘ (:3001); Frontend via ‘npm run dev‘ (:3000).
2. Environment Variables: ‘DATABASE_URL‘ (PostgreSQL 16), ‘JWT_SECRET‘, ‘BACKEND_URL‘ (http://localhost:3001).
3. Unauthenticated Interception: Next.js edge middleware checks ‘zoiko_auth‘ cookie; redirects unauth requests to /login.

STAGE 2: LOGIN, AUTHENTICATION & IDENTITY RESOLUTION

â¢ Page Route: /login (frontend/src/app/login/page.tsx) â¢ API Endpoint: POST /api/v1/auth/login

â¢ Input Fields: Work email (case-insensitive, auto-lowercased), Password (masked with eye toggle), Remember Me.

â¢ Backend Validation: BCrypt verification against identity.local_credentials. Password failures return HTTP 401.

â¢ Cookie Establishment: On 200 OK, sets zoiko_auth, zoiko_shield_access_token (JWT), and zoiko_shield_tenant_id.

â¢ Automatic Redirection: SUPER_ADMIN -> /admin; AUDITOR -> /auditor; TENANT_OWNER / ANALYST -> /customer/dashboard.

STAGE 3: ORGANIZATION REGISTRATION & SOVEREIGN PROVISIONING

â¢ Page Route: /register (frontend/src/app/register/page.tsx) â¢ API Endpoint: POST /api/v1/auth/register-org

â¢ Input Fields: Organization Name, Workspace Slug, Administrator Name, Corporate Email, Master Password, Data Residency Region (us-east-1, eu-west-1, ap-southeast-1), Compliance Framework Packs (SOC2, ISO27001, AI Act).

â¢ Database Execution: Atomically inserts into tenant.tenants, identity.principals, identity.local_credentials, and binds TENANT_OWNER role in authorization.user_roles. Environment is auto-seeded with zero cross-tenant leak.

â¢ Post-Registration: Sets offline credential store in localStorage + cookie for immediate login resiliency after sign-out.

VERIFIED TEST ACCOUNTS & DEFAULT CREDENTIAL DIRECTORY

Email	Password	Role	Landing Route
aditya@zoiko.com	`Aditya@20045678`	TENANT_OWNER	`/customer/dashboard`
sarah.chen@acme.com	`Shield@SecOps2026!`	TENANT_OWNER	`/customer/dashboard`
analyst@zoikoshield-demo.com	`Shield@SecOps2026!`	SECURITY_ANALYST	`/customer/dashboard`
super.admin@zoikoshield.local	`Admin2026!Platform`	SUPER_ADMIN	`/admin`
elena.rostova@auditor.compliance	`Auditor2026!Verify`	AUDITOR	`/auditor`
manualtester2000@zoikoshield.corp	`Password123!`	SECURITY_ANALYST	`/customer/dashboard`

STAGE 4 & 5: CUSTOMER DASHBOARDS & FUNCTIONAL MODULES

THE 3 CUSTOMER DASHBOARDS (/customer/dashboard)

1. Executive View (CISO): Security Posture Score, Critical Exposures, Protection Coverage, Active Incidents.
   Clean database displays clean 0 KPIs. Priority Actions list dynamically binds to open cases in case_management.cases.
2. SOC View (Security Analyst): Triage Pipeline (NEW -> TRIAGED -> ESCALATED), detection rules, and threat feeds.
3. Account View (Compliance/Legal): 7-class data retention policies, Merkle receipts count, and commercial CPQ tier.

THE 14 CUSTOMER SIDEBAR SECTIONS (/customer/[section])

All 14 items dynamically load through CustomerWorkspaceSectionRouter without missing views or mock banners:

1. Security Overview	`/customer/security-overview`	Remediation trends & posture quantification
2. Threats & Incidents	`/customer/threats-incidents`	Case management investigations & timeline audit
3. Exposure Management	`/customer/exposure-management`	Attack surface analysis & unmitigated exposure tracking
4. Assets & Inventory	`/customer/assets-inventory`	Discovered cloud, container & workstation endpoints
5. Vulnerabilities	`/customer/vulnerabilities`	CVE vulnerability register & CVSS severity ranking
6. Detection & Response	`/customer/detection-response`	OCSF detection rule activation & alert trigger count
7. Identity & Access	`/customer/identity-access`	JIT elevation sessions & WebAuthn credential posture
8. Data Security	`/customer/data-security`	Data residency fencing & sensitive DLP exposure audit
9. Cloud Security	`/customer/cloud-security`	AWS, Azure, and Google Cloud security posture evaluation
10. Compliance & Risk	`/customer/compliance-risk`	Automated regulatory framework evaluations (SOC2, ISO)
11. Reports & Analytics	`/customer/reports-analytics`	Cryptographic audit packages & compliance exports
12. Integrations	`/customer/integrations`	Telemetry connector status & ingest latency meters
13. Settings	`/customer/settings`	Workspace identity, tenant region & user profile
14. Command Center	`/customer/dashboard`	Main return link to Executive, SOC, and Account hubs

CORE FUNCTIONAL MODULES DEEP DIVE

Ingestion & Ingest DLQ `shield-ingest (:3002)`
OCSF v1.1 telemetry gateway. Nonce deduplication, HMAC-SHA256 signature verification, and Dead Letter Queue (DLQ) replay.

Alerts & Incident Cases `shield-core (:3001)`
State machine: NEW -> TRIAGED -> CONTAINING -> RESOLVED -> CLOSED. Triage drawer, assignment, and escalation to major case war room.

AI Security Copilot `shield-ai (:3003)`
Hardened LLM investigation. Pre-flight Model Armor prompt sanitization + Spec ´§16.1 10-Field Decision Review Envelope.

SOAR Actions & Quorum `shield-action (:3004)`
R0-R4 authority containment engine. Multi-party dual-custody (4-eyes) approval quorum for destructive actions.

Merkle Evidence Ledger `shield-anchor (:3005)`
RFC 6962 binary Merkle trees + Post-Quantum ML-DSA-65 signatures. Verifiable offline via WebCrypto without data leak.

STEP-BY-STEP QA MANUAL TEST CASES (TC-001 TO TC-015)

TC-001 **Unauthenticated Route Interception** [Role: Unauthenticated] [Route: /customer/dashboard]

Steps: Open browser in Incognito -> navigate to /customer/dashboard.

Expected: Redirected to /login?redirect=%2Fcustomer%2Fdashboard immediately.

TC-002 **Tenant Self-Provisioning & Cell Registration** [Role: New User] [Route: /register]

Steps: Enter Org: Apex Sentinel, Email: admin@apex.io, Pass: Pass123!, Region: eu-west-1 -> Click Submit.

Expected: Receipt displays Sovereign Cell eu-west-1 + HSM hash; auto-logs in to /customer/dashboard.

TC-003 **Login Credential Casing & JWT Issuance** [Role: TENANT_OWNER] [Route: /login]

Steps: Enter Email: Aditya@zoiko.com (mixed case), Password: Aditya@20045678 -> Click Sign In.

Expected: HTTP 200 OK; sets cookies zoiko_auth and shield_access_token; lands on dashboard.

TC-004 **Zero Mock Data Verification (Clean Tenant)** [Role: TENANT_OWNER] [Route: /customer/dashboard]

Steps: Inspect Executive View posture gauge, active incidents count, and priority actions.

Expected: Posture shows 0; Incidents shows 0; displays "No priority actions required at this time".

TC-005 **Complete 14 Customer Sidebar Navigation** [Role: TENANT_OWNER] [Route: /customer/*]

Steps: Sequentially click all 14 sidebar items from Security Overview to Command Center.

Expected: All 14 views render dynamically with zero unhandled exceptions and zero broken routes.

TC-006 **Alert Triage & State Machine Update** [Role: SECURITY_ANALYST] [Route: /alerts]

Steps: Select alert alt-2026-auth-01 -> open drawer -> change status to TRIAGED -> Save.

Expected: Alert status badge turns amber TRIAGED; database alert.alerts.status is updated.

TC-007 **AI Copilot Spec ´§16.1 Decision Envelope** [Role: SECURITY_ANALYST] [Route: /copilot]

Steps: Submit investigation prompt -> inspect generated 10-field review envelope.

Expected: Renders all 10 mandatory fields including Blast Radius, MITRE technique & Attributable Rationale.

TC-008 **SOAR Containment Dual-Custody (4-Eyes)** [Role: ANALYST + OWNER] [Route: /actions]

Steps: Analyst proposes R3 isolation -> Owner reviews and signs approval quorum.

Expected: Action executes only after 2 of 2 distinct signatures are recorded; generates receipt.

TC-009 **WebCrypto Offline Merkle Proof Verification** [Role: Any User / Auditor] [Route: /verify-certificate]

Steps: Paste Merkle receipt JSON -> click "Verify Merkle Inclusion Proof".

Expected: Calculates SHA-256 leaf and root via WebCrypto; displays GREEN valid signature shield.

TC-010 **Non-Destructive DR Restore Drill (G05)** [Role: SUPER_ADMIN] [Route: /admin/platform-health]

Steps: Click "Execute Non-Destructive Restore Drill" -> observe real-time progress bar.

Expected: Steps through 5 stages; generates verified drill receipt confirming zero downtime.

STAGE 6 & 7: RBAC MATRIX & CI QUALITY GATES

ROLE-BASED ACCESS CONTROL (RBAC) PERMISSION MATRIX

Workspace / Route	SECURITY_ANALYST	TENANT_OWNER	SUPER_ADMIN	AUDITOR
Customer Dashboard (/customer/*)	Full Access	Full Access	Full Access	Read-Only
SOC Alert Triage (/alerts)	Full Access	Full Access	Full Access	Denied (403)
AI Copilot (/copilot)	Full Access	Full Access	Full Access	Read-Only
SOAR Containment (/actions)	R0-R1 Only	R0-R3 (Quorum)	Full R0-R4	Denied (403)
Tenant Settings (/admin/tenant)	Denied (Redirect)	Full Access	Full Access	Denied (Redirect)
Platform Health & DR (/admin/platform-health)	Denied (Redirect)	Denied (Redirect)	Full Access	Read-Only
G1 Launch Gate (/admin/g1-gate)	Denied (Redirect)	Denied (Redirect)	Full Access	Read-Only
JIT Break-Glass (/admin/jit-elevation)	Request Only	Quorum Approver	Quorum Approver	Audit Witness
Certificate Verifier (/verify-certificate)	Verified	Verified	Verified	Full Forensic

AUTOMATED CI QUALITY GATES & VERIFICATION HARNESS

Frontend Vitest Full Suite `cd frontend; npx vitest run --maxWorkers=2`
149/149 Passed (100% across 20 files)

Frontend TypeScript Linter `cd frontend; npx tsc --noEmit`
0 Errors, Clean Production Build

Backend Microservices Webpack `cd backend; npm run build`
All 6 microservices compiled successfully

Database Access Policy Check `cd backend; npx jest database-access-policy.spec.ts`
8/8 Passed, 0 cross-module write violations

Ungrounded Terminology Check `cd backend; npm run check:ungrounded-terms`
0 ungrounded terms across 1,841 source files

Public Capability Claims Check `cd backend; npm run check:capability-claims`
All 12 services & 7 domains strictly comply

Controller Access & Swagger `cd backend; npm run access:check; swagger:check`
100% explicit controller access & Swagger coverage

G1 Release Evidence Index `cd backend; npm run check:evidence-index`
All 35 rows checked and verified against artifacts